



Build & Deploy Your Own **MCP** Server

Expose your RAG pipeline as a tool - connect it to any AI agent

Swipe to build it from scratch →



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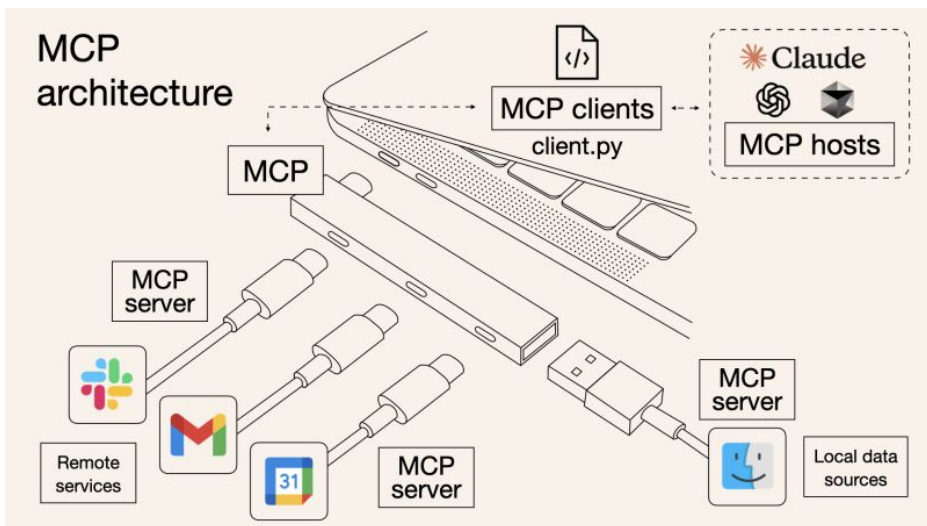
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What is MCP?

MCP (Model Context Protocol) is an open standard that lets AI models talk to external tools and data sources through a unified interface.

💡 Think of it like USB-C - one standard plug that works everywhere.



🔌 3 CORE PIECES

- 🔌 **MCP Server** → exposes your tools
- 🤖 **MCP Client** → the AI agent (Claude, etc.)
- 🔌 **Transport** → how they talk (stdio, SSE, HTTP)





CAPABILITIES

What Can MCP **Expose**?

MCP servers can expose three types of things to an AI agent:

MCP Server

- Tools → functions the LLM can call
 - search_documents(query)
 - ask_rag(question)
- Resources → data the LLM can read
 - file://docs/readme.md
 - db://customers/table
- Prompts → reusable prompt templates
 - summarize_document(doc_id)



For this carousel we focus on Tools - specifically wrapping our RAG pipeline as a callable tool.



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How MCP Works (Flow)

Here's exactly what happens when Claude uses your MCP tool:

User: "What does the doc say about chunking?"



Claude (MCP Client)



1. Sees available tools
2. Decides to call ask_rag()



MCP Server (your code)



3. Receives query
4. Runs RAG pipeline
5. Returns answer



Claude gets result



6. Forms final response

User sees answer



Your job: build steps 3-5. MCP handles everything else.



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Transport Types

Transport = how client and server communicate. Pick based on where you deploy.

stdio (local)

Claude Desktop → stdin/stdout → your script
Best for: local tools, development

SSE (network)

Claude / Agent → HTTP + SSE → your server
Best for: remote servers, APIs

Streamable HTTP (modern)

Agent → HTTP POST → your server
Best for: production, cloud deployments



We'll implement all three. Start with stdio, then upgrade.



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STEP 1

Install FastMCP

FastMCP is the easiest Python library to build MCP servers. One decorator and your function becomes a tool.

```
pip install fastmcp
pip install langchain langchain_community
pip install langchain_huggingface faiss-cpu
pip install langchain_groq
```

✓ **Check it works:**

```
fastmcp version
```

PROJECT STRUCTURE WE'LL BUILD:

```
rag_mcp/
├── rag_pipeline.py    ← your RAG logic
├── server.py          ← MCP server
├── Dockerfile         ← for cloud deploy
└── claude_config.json
```



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</> STEP 2

Build the RAG Pipeline **Module**

First build your RAG pipeline as a standalone module. This is what the MCP tool will call internally.

```
# rag_pipeline.py
from langchain_community.vectorstores import FAISS
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_groq import ChatGroq
from langchain.chains import RetrievalQA

embeddings = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-mpnet-base-v2"
)
vectorstore = FAISS.load_local(
    "faiss_index", embeddings,
    allow_dangerous_deserialization=True
)
llm = ChatGroq(model_name="gemma2-9b-it", temperature=0)

rag = RetrievalQA.from_chain_type(
    llm=llm,
    retriever=vectorstore.as_retriever(search_kwargs={"k": 5}),
    return_source_documents=True
)

def query_rag(question: str) → str:
    result = rag.invoke({"query": question})
    return result["result"]
```



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STEP 3

Create Your MCP Server

Now wrap the RAG pipeline as an MCP tool using FastMCP. One `@mcp.tool()` decorator is all it takes.

```
# server.py
from fastmcp import FastMCP
from rag_pipeline import query_rag

mcp = FastMCP(
    name="RAG Assistant",
    instructions="Use ask_rag to answer questions"
)

@mcp.tool()
def ask_rag(question: str) → str:
    """
    Search the document knowledge base
    and answer a question.
    """
    return query_rag(question)

if __name__ == "__main__":
    mcp.run() # defaults to stdio transport
```



That's your first MCP server. Three lines of real logic.



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✂ STEP 4

Add More Tools

One server can expose multiple tools. Add a search-only tool and a metadata tool alongside ask_rag.

```
@mcp.tool()
def search_docs(query: str, top_k: int = 3) → list[str]:
    """Return raw document chunks relevant to the query."""
    docs = vectorstore.similarity_search(query, k=top_k)
    return [d.page_content for d in docs]

@mcp.tool()
def list_sources() → list[str]:
    """Return all document sources loaded in the knowledge
    base."""
    # pull unique sources from metadata
    docs = vectorstore.docstore._dict.values()
    sources = list({d.metadata.get("source", "unknown")
                    for d in docs})
    return sources
```



Now Claude can choose between: ask a question, get raw chunks, or list sources - depending on what fits the task.



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▶ STEP 5

Run Locally **via stdio**

stdio is the simplest transport. The MCP client launches your script as a subprocess and talks via stdin/stdout.

```
# Test it directly from terminal
python server.py

# Or use FastMCP's built-in dev inspector:
fastmcp dev server.py
```

>_ Output:

```
✓ RAG Assistant MCP server running
📖 Tools available:
  • ask_rag(question)
  • search_docs(query, top_k)
  • list_sources()
🔧 Transport: stdio
```



The inspector opens a browser UI where you can call each tool manually - great for debugging.



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🔌 STEP 6

Connect to Claude Desktop

Claude Desktop reads a config file to know which MCP servers to launch. Add your server there.

```
// ~/Library/Application Support/Claude/claude_desktop_config.json
// (Mac) or %APPDATA%\Claude\claude_desktop_config.json (Windows)

{
  "mcpServers": {
    "rag-assistant": {
      "command": "python",
      "args": ["/full/path/to/rag_mcp/server.py"],
      "env": {
        "GROQ_API_KEY": "your_groq_key"
      }
    }
  }
}
```

🔄 Restart Claude Desktop → you'll see a 🔌 icon → your tools appear. Now Claude can call ask_rag directly in conversation.



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Add SSE Transport (Network)

SSE lets your server run as a web service so any HTTP client - not just Claude Desktop - can connect to it.

```
# server.py - add SSE support
from fastmcp import FastMCP

mcp = FastMCP("RAG Assistant")
# ... same tools as before ...

if __name__ == "__main__":
    import sys
    if "--sse" in sys.argv:
        mcp.run(transport="sse", host="0.0.0.0", port=8000)
    else:
        mcp.run() # stdio for local
```

```
# Run it:
python server.py --sse

# Server live at:
# http://localhost:8000/sse
```



Now any agent that speaks MCP over HTTP can connect to your RAG pipeline.



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STEP 8

Input Validation & Error Handling

Always validate inputs and return clean errors. MCP surfaces these back to the agent so it can retry or explain to the user.

```
from fastmcp import FastMCP
from fastmcp.exceptions import ToolError

@mcp.tool()
def ask_rag(question: str) → str:
    """Answer a question from the knowledge base."""

    if not question or len(question.strip()) < 3:
        raise ToolError("Question too short.")

    if len(question) > 500:
        raise ToolError("Question too long.")

    try:
        return query_rag(question)
    except Exception as e:
        raise ToolError(f"RAG pipeline error: {str(e)}")
```



Claude sees the `ToolError` message and can tell the user exactly what went wrong.



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Dockerize the Server

Package the server into a Docker container for consistent, portable deployment anywhere - local, VPS, or cloud.

```
# Dockerfile
FROM python:3.11-slim

WORKDIR /app
COPY . .

RUN pip install --no-cache-dir fastmcp langchain \
    langchain_community langchain_huggingface \
    faiss-cpu langchain_groq

EXPOSE 8000
CMD ["python", "server.py", "--sse"]
```

```
# Build and run:
docker build -t rag-mcp .

docker run -p 8000:8000 \
    -e GROQ_API_KEY=your_key \
    -v $(pwd)/faiss_index:/app/faiss_index rag-mcp
```

✓ Server is running at <http://localhost:8000/sse>



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Deploy to Cloud (VPS/Railway)

Once Dockerized, deploy to any cloud provider. Here's the pattern using Railway (free tier available).

```
# Option A - Railway CLI
railway login
railway init
railway up

# Option B - Any VPS (DigitalOcean, EC2, etc.)
scp -r rag_mcp/ user@your-server:/app
ssh user@your-server
cd /app && docker compose up -d
```

```
# docker-compose.yml for VPS:
services:
  rag-mcp:
    build: .
    ports:
      - "8000:8000"
    environment:
      - GROQ_API_KEY=${GROQ_API_KEY}
    restart: always
    volumes:
      - ./faiss_index:/app/faiss_index
```



Connect Remote Server to Claude

Point Claude Desktop at your cloud-hosted SSE server instead of a local script.

```
// claude_desktop_config.json
{
  "mcpServers": {
    "rag-assistant-remote": {
      "transport": {
        "type": "sse",
        "url": "https://your-domain.com/sse"
      }
    }
  }
}
```

```
// For protected servers add auth header:
{
  "transport": {
    "type": "sse",
    "url": "https://your-domain.com/sse",
    "headers": {
      "Authorization": "Bearer your_secret_token"
    }
  }
}
```



Restart Claude Desktop - same experience, tools now running in the cloud



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STEP 12

Add a Resource (Bonus)

Beyond tools, expose your document index as a readable resource - agents can browse available docs before deciding what to search.

```
from fastmcp import FastMCP
from fastmcp.resources import Resource

@mcp.resource("docs://index")
def get_doc_index() → str:
    """List all documents in the knowledge base."""
    docs = vectorstore.docstore._dict.values()
    lines = []
    for d in docs:
        src = d.metadata.get("source", "unknown")
        pg = d.metadata.get("page", "-")
        lines.append(f"• {src} (page {pg})")
    return "\n".join(set(lines))
```

"Let me check what documents are available... then I'll search the right one."



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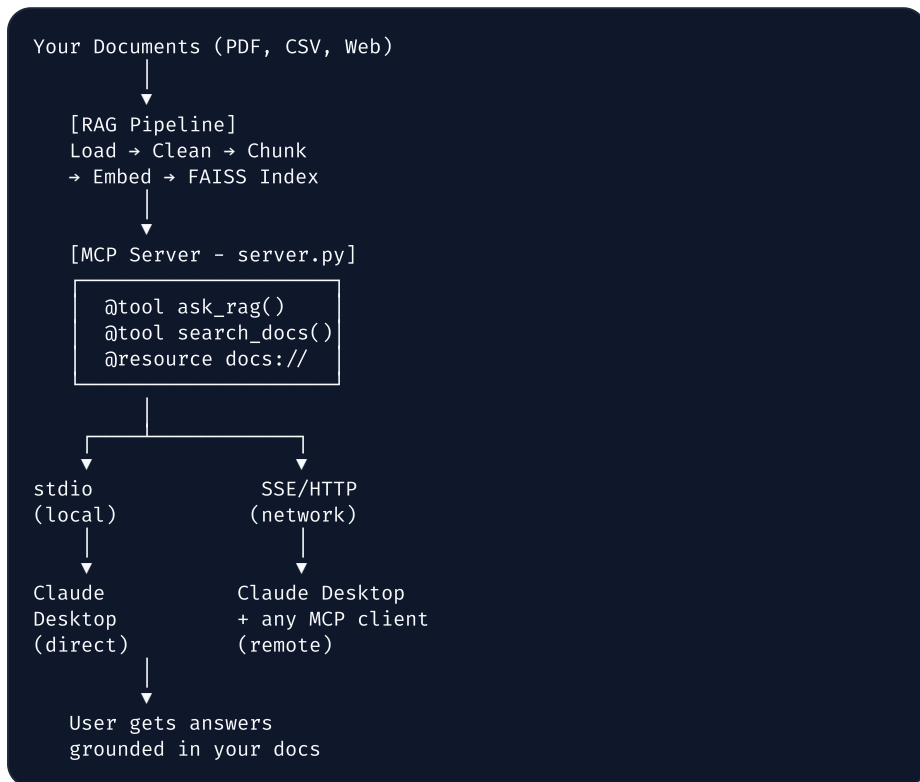


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Full Architecture Diagram

Here's how all the pieces fit together end to end:



Debugging Checklist

When things break, check these in order:

✖ TOOLS NOT SHOWING IN CLAUDE?

- Check config file path is absolute
- Restart Claude Desktop fully
- Run: `fastmcp dev server.py`

✖ RAG RETURNS WRONG ANSWERS?

- Check `faiss_index` path is correct
- Increase `k` (`search_kwargs={"k": 8}`)
- Check chunk size in your pipeline

✖ SSE SERVER NOT REACHABLE?

- Check port 8000 is open in firewall
- Confirm docker container is running
- Check logs: `docker logs rag-mcp`



🚩 CONCLUSION

Summary & What's Next

You now have a complete MCP server that exposes your RAG pipeline as tools - locally and in the cloud.

✅ WHAT YOU BUILT:

- ✓ RAG pipeline module
- ✓ MCP server (3 tools)
- ✓ stdio transport
- ✓ SSE transport
- ✓ Docker container
- ✓ Cloud deployment
- ✓ Error handling

🚀 WHAT TO ADD NEXT:

- API key auth on SSE
- Logging & tracking
- Auto re-index docs
- Conversation memory
- Claude.ai connector

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